Washington College

Department of Environmental Science and Studies

Predicting Stream Discharge Based on Hydrologic Inputs using Machine Learning Methods

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ABSTRACT

Stream discharge is the volume of water flowing through a channel over a specified time period. Stream discharge has major implications for human economy, including hyropower generation and agriculture, and ecological health. Forecasting stream discharge enables educated policy decisions as well as flood and drought warnings. Many variations of process-based and data-driven methods have been developed for hydrologic modeling. We investigate two data-driven methods: Long-Short Term Memory Recurrent Neural Networks and AutoRegressive Integrated Moving Average models. We identify a lack of groundwater inclusion in model development and use these modeling methods to investigate the importance of including groundwater measurements for forecasting in the Susquehanna River Basin. The SRB spans parts of New York, Pennsylvania, and Maryland to provide 50% of Chesapeake Bay's freshwater input. Discharge forecasting is especially important in the SRB because of the watershed's flood-prone status and importance to the Chesapeake Bay. Models were trained on 14 years of USGS groundwater and stream data as well as NOAA weather station data. Limited hyperparamer tuning was done before the models were fit to the training data, cross-validated on twelve 30-day validation splits, and tested against a final 30-day split. Assessed metrics include RMSE, MAPE, KGE, and NSE. The LSTM model without groundwater performed the best across all metrics. Results suggest that groundwater inclusion did not enhance forecasting accuracy.

INTRODUCTION

I. <u>Relevance of Stream Discharge Forecasting</u>

Understanding and forecasting stream discharge has major implications for human safety, economic concerns, and ecological health. Forecasting refers to predicting future events based on past and present data (Middle Atlantic, 2010; Rajib et al., 2025). Short-term stream forecasting is most essential for flood-risk management. Flooding causes between $179.8 and $496.0 billion in damages annually across the United States. Major flooding-related costs include preventative infrastructure upgrades, direct and indirect commercial impact, destruction of commercial physical assets, and damage to homes (Joint Economic Committee, 2024). Allaire (2018) points out that many flood damage estimates often neglect economic loss that is difficult to quantify. While flooding interrupts public services and stalls economic production, low-income households suffer disproportionately. Low-income households often have to divert money away from education and health care to spend on flood recovery and prevention (Allaire, 2018). Forecasting floods can help prevent these costs through preventative actions. Preventative actions include evacuation, transporting portable belongings, stockpiling food and water, and scheduling crop harvesting or planting around forecasted floods. Longer time-frames, such as a 10-day forecast, provide a much greater opportunity for preparedness than shorter, more common 3-day forecasts (Webster & Jian, 2011).

The relationship between forecasting and the economy expands beyond flooding damages. Sophisticated stream discharge forecasts support all water related activities, including hydropower generation, irrigation, and transportation (Girihagama et al., 2022; Fleming & Gupta, 2020). Doyle et al. (2005) explains that discharge is related to many ecological processes. Stream discharge directly affects organic matter transport, algal growth, nutrient retention, macroinvertebrate disturbance, and habitat availability. Discharge can be considered a master variable when analyzing and modeling fluvial ecosystems (Doyle et al., 2005). Confidently

forecasting stream discharge enables well-informed water resource management decisions, as policy makers can stall or enhance pumping based on projected values.

II. Stream Discharge Hydrology

Stream discharge is influenced by many hydrogeologic factors, and forecasting stream discharge must account for these factors. In general, stream discharge is based on precipitation and groundwater recharge inputs (Gao et al, 2025). The relationship between stream discharge, precipitation, and groundwater is based on catchment properties and geomorphology of subsurface pathways (Carlier et al., 2018). During times of low precipitation, a stream is primarily fed by groundwater, which is deemed baseflow (USGS, 2013). As precipitation falls, it runs off the ground surface into the stream, causing the stream's discharge to rise above its base level in a phenomenon called direct runoff. (USGS, 2013). Direct runoff can follow two general pathways: surface runoff or subsurface runoff. Surface runoff does not infiltrate the ground but travels along the ground surface to the stream. Streams that experience a high proportion of surface runoff are flashier, meaning water level changes quickly. Subsurface runoff refers to precipitation that enters the soil and moves towards the stream via subsurface pathways (Carlier et al, 2018). Subsurface geology governs infiltration rates and therefore determines the proportion of surface runoff compared to subsurface runoff (Carlier et al., 2018). The dynamics of runoff pathways are influenced by vegetation, soil, topography, and geology. These fundamental characteristics of a catchment basin affect infiltration, evapotranspiration, interception, and overland and subsurface flow (Carlier et al., 2018). Many processes influence how much water input the stream receives as well as when the input reaches the stream.

Accounting for this multitude of factors in stream discharge modeling has traditionally required extensive mathematical equations.

III. Classical Stream Discharge Modeling

The basis of hydrologic modeling lies with Henry Darcy's sand filter experiments, through which Darcy observed and quantified how water flows through porous mediums. Darcy's equations govern important groundwater properties, including average linear flow and porous flow versus conduit flow (Darcy, 1856; Ritzi & Bobeck, 2008). Since Darcy's experiments, many process-based models have been developed. Girihagama et al. (2022) categorizes process-based models as conceptual lumped models and physical process based distributed models. Conceptual lumped models simplify the physical properties involved in runoff generation through mathematical formulas. The simplifications make conceptual lumped models easier to use and are based on sound principles, but these models do not capture spatial variability of hydrologic processes. Physical process based distributed models capture the spatial variability of the watershed through tailored equations. This watershed-specific tailoring causes physical process based distributed models to not be transferable between watersheds (Girihagama et al., 2022). The process-based modeling approach has been applied successfully and is theoretically intact. However, process-based modeling struggles with constraints such as oversimplified parameterization, extensive data requirements, and high computational complexity (Hamzeh et al., 2025). With recent expansions in data collection, data-driven modeling approaches stand as a valuable alternative to process-based modeling.

IV. Data-driven Discharge Modeling

Data-driven models refer to algorithms that learn patterns from existing data. Their main advantage over process-based modeling is that data-driven models do not require detailed knowledge of a stream system's hydrogeologic properties (Geer, 2021). Also, data-driven models can run with less input data, which has led to an increase in their popularity (Hamzeh et al., 2025). Data-driven models encapsulate both traditional statistical models as well as deep network machine learning (ML) techniques (Girihagama et al., 2022; Hamzeh et al., 2025; Xu & Liang, 2021). Statistical approaches quantify the relationship between input and output variables. The most powerful statistical models for hydrology are those that incorporate time-series analysis, such as Autoregressive Integrated Moving Average (ARIMA) models (Girihagama et al., 2022; Hamzeh et al, 2025). ARIMA models rely on long records of historical data to derive relationships among data sequences and forecast future values (Girihagama et al, 2022). An advanced version of ARIMA is SARIMAX, which allows for the inclusion of seasonal trends (S) and the input of external independent variables (X). In hydrology, SARIMAX models enable the inclusion of relevant exogenous information, such as precipitation and temperature (Perazzollo et al., 2025). The SARIMAX model forecasts future values based on past values of the target variable as well as coefficient values of exogenous input data. The parameter weights allow us to draw some conclusions about a system's hydrogeologic properties. For example, if a SARIMAX model weighs precipitation very strongly relative to groundwater, that would indicate the stream relies on meteorological input moreso than groundwater baseflow. Alternatively, deep network machine learning models process data and extract non-linear relationships without explicitly defining physical properties. The ability of ML models to learn patterns without defining physical properties improves the model's generalizability to other watersheds, a feature that process-based models and statistical models struggle with (Girihagama et al., 2022).

Deep network machine learning models share a history with hydrologic modeling beginning in the 1990s. The first deep network models were Artificial Neural Networks (ANNs). ANNs were inspired by biological cognitive processes and consist of layers of units, or neurons. The basic ANN architecture includes an input layer, one or more hidden layers, and an output layer. Information gets passed from the input layer to the hidden layers, and each neuron in the first hidden layer computes a single output based on the inputs. The neuron outputs then get passed to each neuron in proceeding hidden layers until the output layer produces a comprehensible prediction value. ANNs learn through a backpropagation procedure, where the model calculates a loss function based on its prediction error and adjusts neuron weights via a backward pass through the layers (Xu & Liang, 2021). While ANNs are powerful tools to process and learn from complex data, the original ANN architecture struggles with time-dependent hydrology data.

The Long-Short-Term-Memory (LSTM) variant of the Recurrent Neural Network (RNNs) architecture has been more successful in hydrologic modeling (Cheng et al, 2020; Girihagama et al, 2022; Hamzeh et al, 2025; Xu & Liang, 2021). RNNs have the capability to process variable input and output lengths while keeping data in sequential order. These features are a benefit of RNNs over ANNs for time-dependent hydrologic forecasting, but traditional RNNs have difficulty retaining information from much earlier sequences (Girihagama et al., 2022). LSTM is a modification to the RNN model that allows for capturing long-range relationships. LSTMs have been used in hydrologic applications for rainfall-runoff modeling, stream discharge predictions for ungauged basins, modelling flash flood events and flood forecasting, and simulating reservoir outflows (Girihagama et al., 2022).

V. Application to Susquehanna River Basin

Within the literature around stream discharge forecasting, many researchers develop models without accounting for groundwater. Galavi et al (2023) develop an ARIMA model which does not use any exogenous input variables and find it to be ineffective at forecasting water level for long time horizons. Hamzeh et al (2025) compare different types of forecasting models using past stream gauge and upstream gauge data. They acknowledge that discharge forecasts have long-term dependencies based on precipitation runoff; this long-term impact of precipitation may be captured statistically by groundwater level. Gao et al. (2025) studied the Yanglou catchment in eastern China and found that stream discharge is associated with periods of rapid groundwater rise. High soil permeability in the Yanglou catchment suggests that overland flow is not the primary runoff transport pathway. The research in the Yanglou catchment suggests that a stream's response to groundwater levels is based on the underlying geology of the catchment. Randall's (2010) research in the northern Susquehanna River basin (SRB) similarly discusses the relationship between groundwater and stream discharge. During periods of low flow, groundwater from sand and gravel deposits is the principal source of stream discharge. Many sites in the northern SRB are incised in impermeable till or bedrock, which prevent groundwater from flowing anywhere except into the stream (Randall, 2010). This suggests that stream discharge forecasts of the northern SRB would highly rely on groundwater levels. While the New York section of the SRB has been researched in detail, the Pennsylvania and Maryland portions lack descriptions of groundwater hydrology.

VI. Hypothesis

By building forecasting models for the SRB, we can gain insight into its localized hydrogeologic characteristics. For each study site, we will build four models. We will train two ARIMAX and two LSTM models to forecast future stream discharge values based on hydrologic

and meteorologic inputs. We aim to study the impact of groundwater data on discharge forecasting. As such, one ARIMAX and one LSTM model for each site will not include groundwater level measurements to act as controls to compare the groundwater-included models against. We will develop both ARIMAX and LSTM models because ARIMAX has more interpretable parameter weights, while LSTM models are more commonly used in other hydrology modeling research. We hypothesize that both ARIMAX and LSTM models that include groundwater as an input will forecast stream discharge of SRB tributaries more accurately than models that do not include groundwater. Forecasting accuracy will be measured by root mean squared error (RMSE), Mean Absolute Percent Error (MAPE), the Kling-Gupta Efficiency (KGE), and the Nash-Sutcliffe Efficiency (NSE).

METHODS

I. Susquehanna River Basin, Site Selection

The Susquehanna River Basin (SRB) is an ecologically and economically valuable watershed that spans through upstate New York, most of eastern Pennsylvania, and a small section of Maryland. The Susquehanna drains into the Chesapeake Bay, the United States' largest and most economically valuable estuary. The SRB provides 50% of the Chesapeake's freshwater input, so understanding the SRB's discharge directly relates to the Chesapeake Bay (Leathers et al, 2018). The SRB's headwaters in New York and Pennsylvania were carved by glacial movements and deposits. Glacial action created many bedrock and valley-fill aquifers, which are valuable human water resources (Heisig & Fleisher, 2022). The SRB is a particularly effective area for investigating the role of groundwater in forecasting models because sand-and-gravel aquifers provide groundwater discharge during low flow periods (Randall, 2010)

Selecting sites in the SRB depended on monitoring instrument placement and data availability. That is, three instruments needed to be in close proximity: a USGS stream guage, a USGS groundwater well, and a NOAA weather station. Additionally, we preferred sites with extensive data records so as to provide the models with more information to train on.

Site A is located in and around Biglerville, Pennsylvania. The stream guage is USGS-01571184 at coordinates (40.0309245327505, -77.3085949472522). The groundwater well is USGS-400209077183301 at coordinates (40.0352777777778, -77.3080833333333). The weather station is USC00360656 at coordinates (39.93468, -77.25459).

Site B is located in and around Lewistown, Pennsylvania. The stream guage is USGS-01567000 at (40.4784167735445, -77.1291483971583). The groundwater well is USGS-402735077100901 at (40.4604444444444, -77.1686944444445). The weather station is USC00364992 at (40.5864, -77.5692).

Site C is located around Williamsport, PA. The stream guage is USGS-01550000 at (41.418058820116904, -77.0329971470846). The groundwater well is USGS-420710077052101 at (41.4079166666667, -76.99475). The weather station is USW00014778 at (41.24295, -76.92173).

II. Code Development

All code functions were developed in Python 3.12 using Jupyter Notebooks, GitHub, and GitHub Codespaces. I used Jupyter Notebooks to experiment with model design as well as familiarize myself with the USGS and NOAA data portals. Then, I converted the code into GitHub and GitHub Codespaces for final edits, running, and sharability. Required external packages included pandas, numpy, plotly, matplotlib, seaborn, sktime, scikit-learn, dataretrieval, pmdarima, and permetrics.

USGS data included two key parameters: stream discharge and groundwater depth below ground surface. These were accessed using the *dataretreival* package managed by USGS. The stream discharge data used data code 00060, or the daily mean discharge in cubic feet per second. The groundwater level used code 72019, which often returned three values for each day: maximum, minimum, and mean. I only selected the mean groundwater level and created a data frame with date, groundwater level, and stream discharge columns.

Because statistical models train more accurately with normalized data, I included a function that checked the skew value of groundwater level and stream discharge (Geer 2021). If the skew for either variable was greater than 0.75, the variable underwent lofarithmic transformation. In the case of a negative skew and skew less than -0.75, which was often the case with groundwater level, the values were reflected before logarithmic transformation. For groundwater, this changed the meaning of the variable from "groundwater level in feet below ground surface" to "groundwater level in feet above historic groundwater minimum."

For weather data access, I explored the NOAA weather station site and selected the closest weather station to the stream and groundwater monitors that had a sufficient data record. I downloaded data for each weather station and read it into the program, selecting for the following daily variables: maximum temperature, precipitation, snow, and snow depth. The program selected weather data for the given start and end dates. Then, the program merged weather data and hydrology data into one dataframe. Because the statistical models cannot run with missing data values, I implemented an interpolation step. Missing values were first interpolated linearly. That is, if one data point was missing between two known data points, then missing data point became the mean of its two neighbors. Also, the data were backfilled and

forward-filled. If data were missing at the start or end of the dataframe, the missing values were set equal to the first or last actual data point, respectively.

Following the dataframe setup and cleaning, the data was prepared for the models. I created a copy of the dataframe excluding the groundwater level column, thus establishing a groundwater-included dataframe and a groundwater-absent dataframe. Each dataframe underwent identical processes: splitting into training, validation, and test sets. Then, I fit each split dataframe with an LSTM model and with a SARIMAX model. The test data was split from the rest of the data by separating the last 30 days in the dataset. The test sets were not seen by the model during training, but were left for calculating final forecast performance metrics. For actual model tuning, the training/validation data was split using an expanding window splitter method. The windows were each 30 days in length, and there were 12 splits. Therefore, the first training fold spanned frome the established start data (January 1st, 2010) to 360 days before the final point in the training/validation set. Each model trained on this first training set, T_1, established some parameter values, and scored its parameter values against the first validation set, V_1, or the next 30 days of data. Then, the model moved onto the next training and validation sets, T_2 and V_2. T_2 can be thought of as $T_1 + V_1$. So, the second training set includes all of the points in the first training set and all of the points in the first validation set. V_2 is the next set of 30 days after the end of T_2. This pattern repeats itself so that the model trains against all seasons in a year.

III. <u>LSTM</u>

The LSTM was set up using the *sktime* NeuralForecastLSTM() function (Sktime, 2025). The function required specified hyperparameters: a minmax scaler, an input size of 365, a batch size of 32, a learning rate of 0.005, and 200 maximum steps, 2 hidden layers, and 128 units per layer. The minmax scaler sets all input variables on a range between 0 and 1, where 0 is the

minimum value of that variable, and 1 is the maximum value; all other values are set proportionally between the minimum and maximum. The minmax scaler is more sensitive to outliers which is important for capturing occasional high discharge events (Amorim at al., 2022). The input size of 365 lets the LSTM look back at the past year of data when forecasting a future point. Limiting the LSTM's exposure to past data helps prevent long model training times and enables the LSTM to learn new data patterns instead of relying too strongly on annual cycles. Allowing for 365 days of input still allows the model to pick up some annual signal. Batch size is the number of data points the model looks at before adjusting its parameter values. A batch size of 32 is relatively small, which couples well with the low learning rate of 0.005. The small batch size means that the LSTM adjusts its parameters based on a small, potentially very noisy subset of data, but the low learning rate prevents the parameter values from adjusting too drastically. This combination helps capture the variability in hydrology data. The 200 maximum steps means the LSTM passes over 200 batches of data, which allows the model the opportunity to adjust its parameter values while keeping computational runtime low. Finally, the 2 hidden layers of 128 units help the model capture complex relationships without being overly complex, so as to prevent overfitting.

The LSTM trained its parameters against the 12 cross-validation splits. Each fit was scored using root mean squared error (RMSE), and the model selected the best model weights.The selected LSTM forecasted the final 30 days of data using estimates for future predictor variables. These estimates were calculated as historical averages for each parameter value by calendar date, and this forecast was scored against the 30-day test set.

IV. SARIMAX

To set up the SARIMAX model, I relied on the sktime AutoARIMA function, which is built on pmdarima. These ARIMA models required the hyperparameters: p, d, and q as well as seasonal versions P, D, and Q. Other hyperparameters were period of seasonal differencing, model evaluation information criterion, and inclusion of stepwise fitting algorithm. Lowercase p is the autoregressive term, which adds a weight based on a recent data point. The maximum p allowed was 2, meaning the model could use up to the past 2 data points as a basis for the next forecast. To avoid high computational costs, maximum seasonal P was set to 0. Lowercase d allows for differencing between points to keep the data stationary. Maximum lowercase d was set to 1 and maximum seasonal D was set to 1. These differencing values allow the model to remove trend and seasonality to predict values more consistently. Lowercase q is the moving average term, which allows the model to use the error in the most recent prediction to forecast the next prediction value. Maximum lowercase q was set to 2, and maximum seasonal Q was set to 0 to avoid high computational costs. The period of seasonal differencing was set to 365, to indicate 365 days pass before a seasonal pattern repeats. The model selector was set to AICc, or Akaike Information Criterion corrected for small sample size. The stepwise fitting algorithm was set to true, which speeds up the model fitting process and is less likely to overfit than other fitting algorithms.

Cross-validation for SARIMAX was purely informational. The model ran for each of the cross validation folds, and for each fit, produced an RMSE score. This allows us to view how the SARIMAX performs across different seasons as opposed to solely the test set. Then, the SARIMAX was refitted on all of the training and validation data. The SARIMAX model

predicted the test set for 30 days by relying on daily historical average values as estimated for predictor variables.

V. Model Scoring

Each site had 4 sets of scoring metrics. The scoring metrics calculated included Root Mean Squared Error (RMSE), Mean Absolute Percent Error (MAPE), Nash-Sutcliffe Efficiency (NSE), and Kash-Gupta Efficiency (KGE). For each site, these metrics were calculated for the groundwater-included LSTM, groundwater-included ARIMAX, groundwater-absent LSTM, and groundwater-absent ARIMAX. The scoring metrics provide an estimate on overall performance of a site's forecast per model. To assess the ability of each model setup to forecast within the SRB generally, the performance metrics were averaged across all sites.

The ARIMAX model is especially important when assessing the contribution of groundwater to model performance because ARIMAX models include parameter weights for predictor variables. It will be noted if the ARIMAX model adds significant weight to the groundwater level as a predictor variable.

An important aspect of hydrologic forecasts is accuracy as time into the future increases, or avoidance of forecast degradation. To measure this, we calculated the point-wise absolute difference between forecasted values and actual values. Therefore, each model for each site has an array of 30 absolute difference values. These were also averaged across sites, to produce a point-wise mean absolute difference per model type. A rate of change and R^2 value were calculated for each model's mean absolute difference set. A positive rate of change would indicate forecast reliability decreases further into the future.

RESULTS

The mean performance metrics for each model type are as follows (Table 1). For the groundwater-included LSTM, RMSE = 0.94, MAPE = 0.32, KGE = -0.66, and NSE = -5.21. For the groundwater-absent LSTM, RMSE = 0.52, MAPE = 0.10, KGE = -0.08,and NSE = -0.13. For the groundwater-included ARIMA, RMSE = 0.80, MAPE = 0.20, KGE = -0.21, and NSE = -2.28. For the groundwater-absent ARIMA, RMSE = 0.69, MAPE = 0.13, KGE = -0.35, and NSE = -0.93.

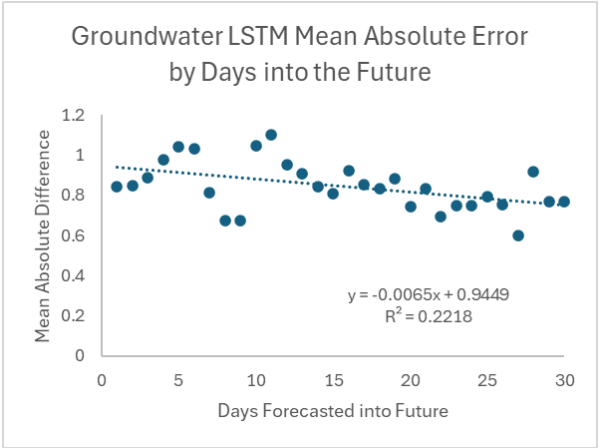
Table 1: Mean performance metrics for each model type, with each value representing the mean across all SRB sites (n=3). The non-groundwater LSTM has the lowest RMSE and MAPE and the greatest KGE and NSE. RMSE = Root Mean Squared Error; MAPE = Mean Absolute Percent Error; KGE = Kling-Gupta Efficiency; NSE = Nash-Sutcliffe Efficiency.

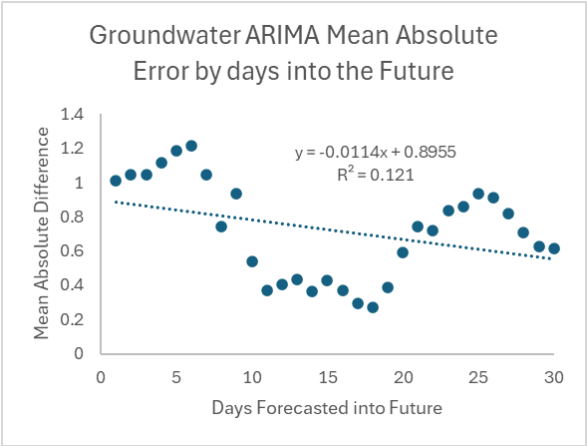
Model Type	**RMSE**	**MAPE**	**KGE**	**NSE**
LSTM GW	0.94	0.32	-0.66	-5.21
LSTM non-GW	0.52	0.10	-0.08	-0.13
ARIMA GW	0.80	0.20	-0.21	-2.28
ARIMA non-GW	0.69	0.13	-0.35	-0.93

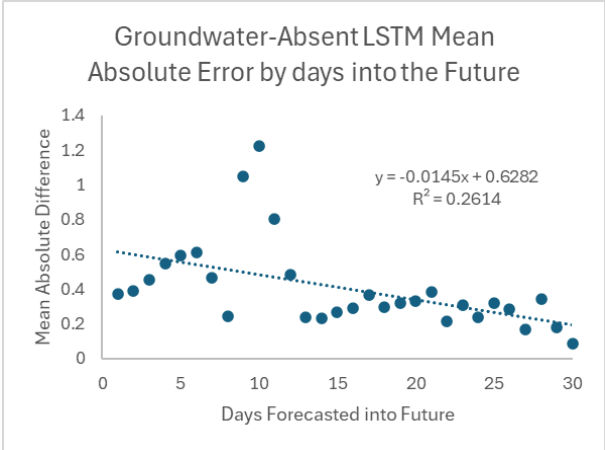
The forecast degradation determination produced a set of mean absolute differences between the forecasted and actual stream discharge values. A set of this type existed for each model. For groundwater-included LSTM, the rate-or-change was -0.0065 with an R^2 of 0.2218

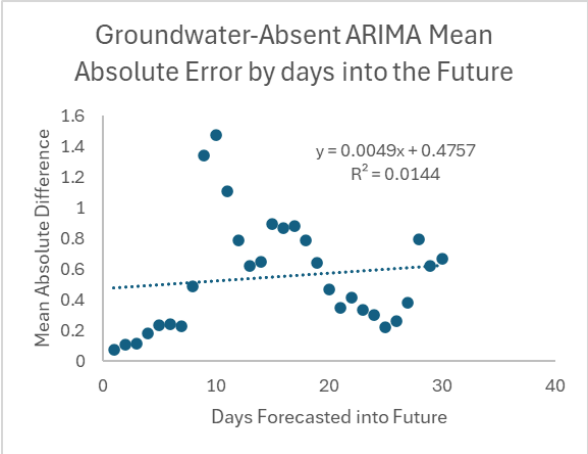
(Figure 1). For groundwater-absent LSTM, the rate-of-change was -0.0114 with an R^2 of 0.121 (Figure 2). For groundwater-included ARIMA, the rate-of-change was -0.0145 with an R^2 of 0.2614 (Figure 3). For groundwater-absent ARIMA, the rate-of-change was 0.0049 with an R^2 of 0.0144 (Figure 4).

All ARIMA models had a non-zero and statistically significant parameter weight for groundwater level. For site A, the groundwater coefficient was -0.1745 with a p-value<0.001. For site B, the groundwater coefficient was -0.4918 with a p-value<0.001. For site C, the groundwater coefficient was -0.5104 with a p-value<0.001.









Figures 1-4: Mean Absolute Difference values between forecasted stream discharge and actual discharge, calculated for each day into the future. Equations on each plot display the positive or negative trend of forecast reliability as forecast horizon increases. The R^2 values indicate how strong the relationship is between days into the future and forecast accuracy.

DISCUSSION

I. Relevance of Groundwater

The presence or absence of groundwater data in these models corresponds to an observable difference in model performance.The non-groundwater LSTM and ARIMA models have lower RMSE and MAPE values than their groundwater-included counterparts. In other words, the non-groundwater models predict stream discharge values closer to the real values compared to the groundwater-absent models. The decision to use historic daily average groundwater values likely influenced these forecasts. For sites A and B, historic groundwater influenced the model to forecast much higher values, which led to generally poorer fits. The same effect happened with site C, but C experienced a rapid and consistent increase in real discharge (figure 5). Site C's ARIMA without groundwater model forecasted a more typical, low precipitation month and underestimated discharge, thereby leading to a low KGE value. This example calls attention to the important interplay between precipitation and groundwater. For all sites, groundwater level was high during the test month but few precipitation events occurred with sites A and B. At site C, when real precipitation and real groundwater level more closely align with historic averages, the groundwater-included ARIMA model performs relatively well.

Aside from overall model scores, the ARIMA model's groundwater coefficients suggest that groundwater is very important to actual stream discharge. All of the groundwater coefficients were on the same order of magnitude as the ARIMA terms, so fluctuations in groundwater level had a meaningful impact on predicted values Additionally, each ARIMA model calculated the groundwater coefficient as having a p-value<0.001. The poor model performance, therefore, may be because the 30-day test period (December 3, 2024 – January 1, 2025) had extreme qualities. In further iterations of this question, it would be beneficial to set the

models' test split as another month of the year. It is possible that December 2024 had unusual weather for the entire SRB region, and the performance metrics, therefore, may not represent the models' performances for forecasting other seasons of the year.

II. Model Applicability

An important feature of the trained models is their ability to maintain a consistent forecast across the forecast horizon. Many forecasting models operate on a horizon of 1 to 10 days (Galavi et al, 2013; Girihagama et al, 2022; MARFC, 2010; Webster & Jian, 2011). Models optimized for small horizon forecasting often produce more accurate predictions than our models for the specific, small horizon. Developing a small horizon flood forecaster has value for emergency flood preparedness, but pre-flood measures are limited by the amount of advance warning time. Webster & Jian (2011) acknowledge that rural, developing communities benefit from a forecast horizon of ten days. Farmers can use the ten days to relocate livestock and to reorganize their planting and harvesting schedule to avoid ruined crops. The LSTM and ARIMA models here were trained and validated to produce a low RMSE across the entire thirty-day horizon. While the models are not particularly effective at forecasting in the short term and generally fail to capture extreme events, their absolute error stays consistent throughout the thirty days.

III. Comparisons to Other Model

Comparing models across research is made difficult because RMSE, KGE, and NSE are dependent on the scale of the data. However, MAPE is an unbiased statistic where a smaller value indicates a better predictability of the model (Galavi et al, 2013). Galavi et al (2013) built ARIMA and Adaptive Neuro-Fuzzy Infernce System (ANFIS) models to forecast one-day-ahead water level in the Klang River. Their final ARIMA and ANFIS models had MAPE values of

0.337 and 0.404, respectively. All versions of our model have a lower MAPE than Galavi et al, suggesting that our parameter selection and hyperparameter tuning produced relatively effective machine learning models.

CONCLUSION

The ARIMA and LSTM models forecast general 30-day trends effectively. All versions of our models struggle to capture extreme discharge fluctuations, which is likely a cause of model scoring during our validation steps. The groundwater-absent LSTM could be used to forecast general 30-day discharge for ecological modeling purposes, but these models should not be used for flood warning. Most of our results suggest the groundwater did not improve model forecast quality. However, there are alternative methods for including groundwater in discharge forecasting. For example, groundwater could also be forecasted using an ARIMA model, and the results of that forecast can be an input into the primary discharge forecast model.

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APPENDIX

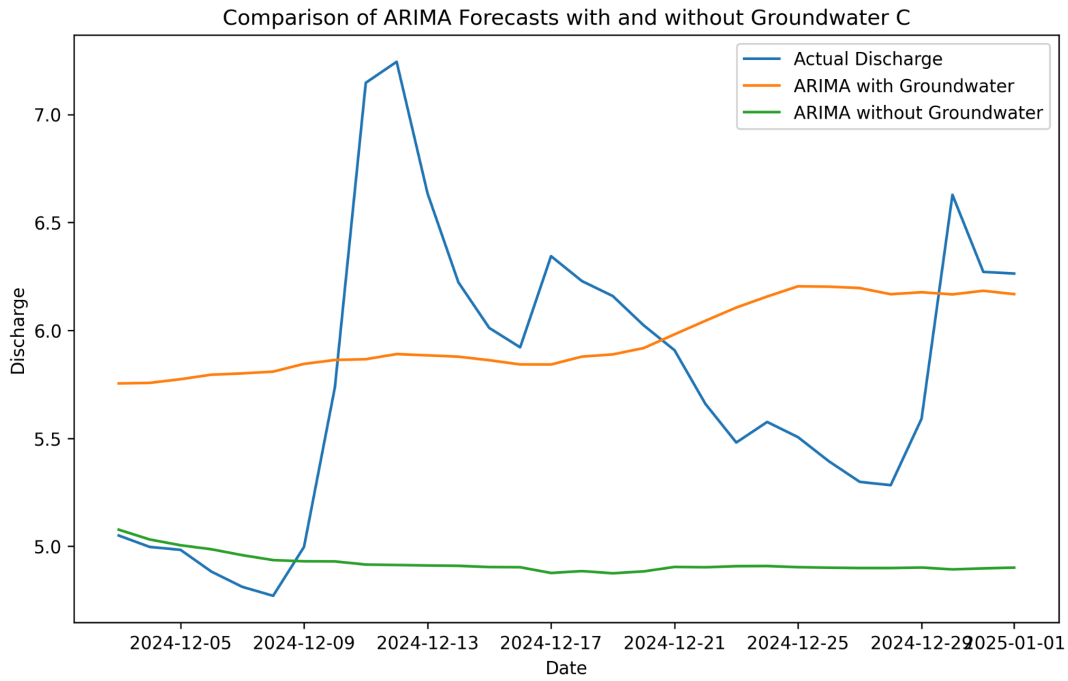

Figure 5: Sample plot of actual discharge compared to the ARIMA forecasting models. The ARIMA groundwater-included model performed much better by forecasting higher discharge than the groundwater-absent model.